

ELEC5803 HLS Exam Study

Step-by-Step Solution to Question 5

Course-slide basis

This solution follows the redundancy and reliability discussion in `Slides/ELEC5803_Slides10_Testability and Dependability_AA2.pdf`, especially the matrix-multiplication example, DMR/TMR/NMR, temporal redundancy, and information redundancy sections.

1. Interpreting the problem

The matrices have dimensions

$$A \in \mathbb{R}^{3 \times 4}, \quad B \in \mathbb{R}^{4 \times 3}, \quad C = AB \in \mathbb{R}^{3 \times 3}.$$

So the output matrix has

$$3 \times 3 = 9$$

elements. The question explicitly says:

- each output element computation is treated as one *operation*,
- the MAC is the computation module,
- the fault process follows a Poisson model with average rate λ .

Therefore, if one operation takes time t , then from the slide figure:

$$P_{\text{ne}}(t) = e^{-\lambda t}, \quad P_{\text{error}}(t) = 1 - e^{-\lambda t}.$$

For compactness, define

$$q = P_{\text{ne}}(t) = e^{-\lambda t}, \quad p = P_{\text{error}}(t) = 1 - e^{-\lambda t}.$$

2. Part (a): DMR and TMR

2.1 Baseline single-module system

With one MAC module, the probability that an operation finishes without error is

$$q = e^{-\lambda t},$$

and the probability that the operation is erroneous is

$$p = 1 - e^{-\lambda t}.$$

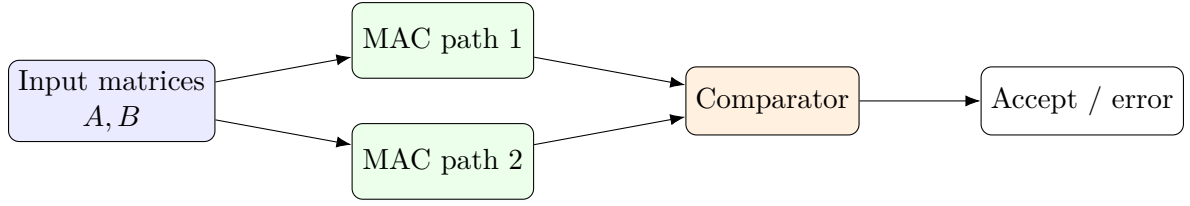


Figure 1: DMR for one matrix-element computation.

2.2 Dual Modular Redundancy (DMR)

DMR duplicates the computation and compares the two outputs.

DMR is mainly an **error-detection** scheme. If at least one module is faulty, the DMR system is considered to have encountered an error/failure event for that computation.

Hence the probability of DMR error is:

$$P_{\text{DMR,error}} = 1 - q^2 = 1 - e^{-2\lambda t}.$$

Equivalently,

$$P_{\text{DMR,error}} = 2pq + p^2.$$

Interpretation:

- $2pq$: exactly one module is erroneous,
- p^2 : both modules are erroneous.

2.3 Triple Modular Redundancy (TMR)

TMR uses three replicated modules and a 2-out-of-3 majority voter.

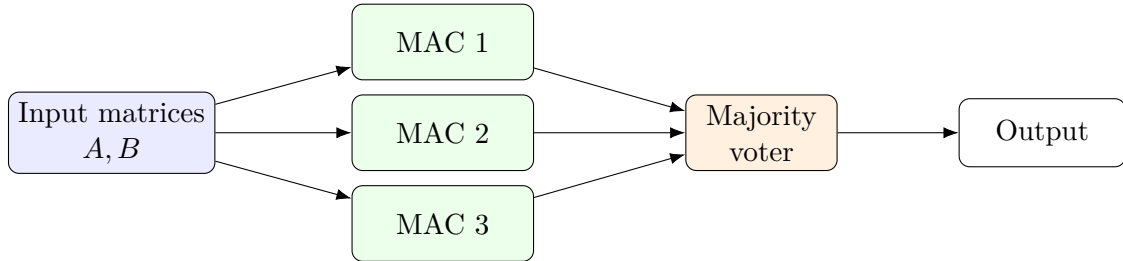


Figure 2: TMR for one matrix-element computation.

From the slides, the TMR error probability is

$$P_{\text{TMR,error}} = 3p^2q + p^3.$$

Why? The TMR output is wrong only if:

- any two of the three modules are erroneous, or
- all three modules are erroneous.

So

$$P_{\text{TMR,error}} = \binom{3}{2}p^2q + \binom{3}{3}p^3 = 3p^2q + p^3.$$

Substituting $p = 1 - e^{-\lambda t}$ and $q = e^{-\lambda t}$:

$$P_{\text{TMR,error}} = 3(1 - e^{-\lambda t})^2e^{-\lambda t} + (1 - e^{-\lambda t})^3.$$

2.4 Final answer for part (a)

$$P_{\text{DMR,error}} = 2pq + p^2 = 1 - e^{-2\lambda t}$$

$$P_{\text{TMR,error}} = 3p^2q + p^3$$

with

$$p = 1 - e^{-\lambda t}, \quad q = e^{-\lambda t}.$$

3. Part (b): NMR, N-temporal redundancy, and information redundancy

3.1 N-Modular Redundancy (NMR)

This generalizes TMR to N replicated modules with a k -out-of- N voter.

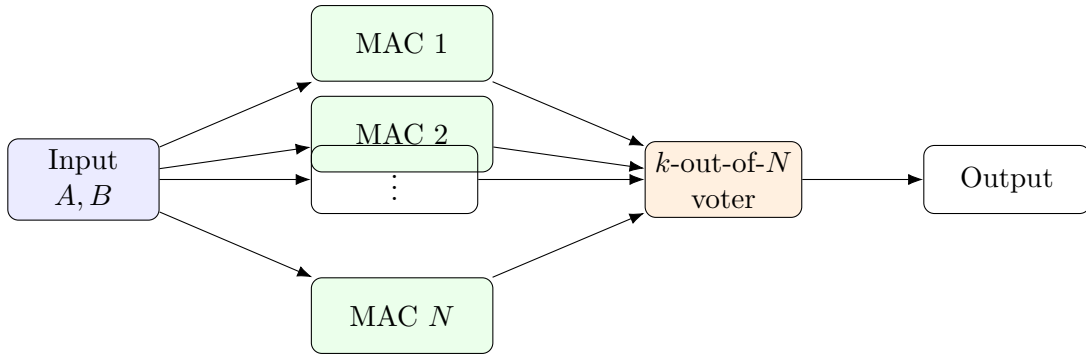


Figure 3: Generalized NMR (k -out-of- N) architecture.

If the system is correct when at least k modules are fault-free, then the error probability is:

$$P_{\text{NMR,error}} = \sum_{i=N-k+1}^N \binom{N}{i} p^i q^{N-i}.$$

For TMR, $N = 3$ and $k = 2$, which reduces to $3p^2q + p^3$.

3.2 N-Temporal redundancy

Temporal redundancy reuses the same hardware N times over time instead of replicating hardware spatially.

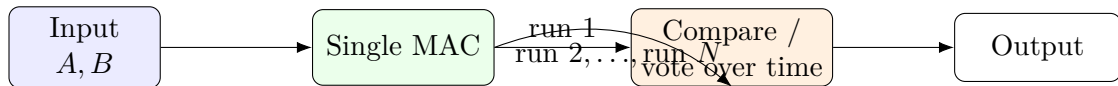


Figure 4: N-temporal redundancy: repeated evaluation on the same MAC hardware.

Here:

- area overhead is small,
- time overhead is high because the same operation is repeated N times,
- timing becomes less deterministic, exactly as noted in the slides.

3.3 Information redundancy

Information redundancy augments the operands with checksums.

For the present 3×4 by 4×3 multiplication:

- original output size: $3 \times 3 = 9$ operations,
- checksum-augmented output size: $(3 + 1) \times (3 + 1) = 4 \times 4 = 16$ operations.

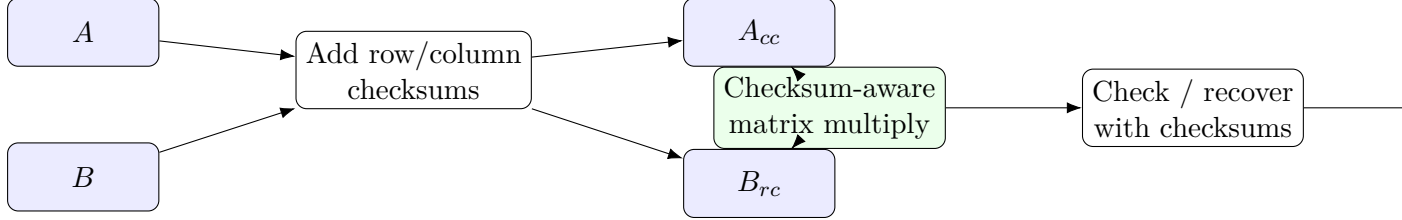


Figure 5: Information redundancy by checksum augmentation.

3.4 Comparison table

Method	Area cost	Time cost	Power / energy trend
NMR	High; about $N \times$ compute hardware plus voter	Low extra time; usually near-baseline latency with small voting overhead	High instantaneous power and thermal stress due to parallel replicas
N-temporal redundancy	Low area; one compute module reused	High; roughly $N \times$ execution time	Lower instantaneous power than NMR, but higher total energy because work is repeated
Information redundancy	Moderate; checksum logic and extra storage	Deterministic overhead; for this case 16 operations instead of 9	Moderate increase because extra checksum operations are computed

3.5 Short discussion

This matches the slides:

- **spatial redundancy** improves functional reliability but increases area, power, and thermal stress,
- **temporal redundancy** saves area but hurts timing reliability because execution time increases,
- **information redundancy** adds deterministic computational overhead through checksum processing.

4. Part (c): combining methods for lower cost and higher reliability

The best practical design is usually *hybrid*, not purely one technique.

4.1 Reasonable hybrid strategy

A low-cost and reliable redesign for the matrix multiplier could be:

- use **information redundancy** (checksums) globally to detect most computation errors at moderate cost,
- use **selective TMR** only for the most critical blocks, such as the voter, control logic, or the MAC datapath if the application is safety-critical,
- use **temporal re-execution** only after checksum mismatch is detected, instead of repeating every computation all the time,
- optionally keep a **cold spare** MAC for recovery without permanently paying full thermal cost.

4.2 Why this is better

This hybrid approach reduces cost because:

- full NMR everywhere is expensive in area and power,
- pure temporal redundancy is too slow,
- pure information redundancy detects well but does not always mask errors by itself.

By combining them:

- checksums provide cheap error detection,
- selective spatial redundancy protects only the highest-risk logic,
- temporal retry is activated only when needed.

So the resulting system achieves:

- better functional reliability than using only one light-weight method,
- lower area and power than full TMR/NMR everywhere,
- lower average execution-time penalty than always-on temporal redundancy.

5. Final exam-style answers

(a)

Let

$$p = 1 - e^{-\lambda t}, \quad q = e^{-\lambda t}.$$

Then:

$$P_{\text{DMR,error}} = 2pq + p^2 = 1 - e^{-2\lambda t}$$

$$P_{\text{TMR,error}} = 3p^2q + p^3$$

(b)

- NMR: replicate hardware spatially and use a k -out-of- N voter.
- N-temporal redundancy: repeat the same computation N times on the same MAC.
- Information redundancy: augment matrices with row/column checksums and verify the output.

For the given case, information redundancy increases the number of matrix-element operations from 9 to 16.

(c)

Use a hybrid design:

- checksum-based information redundancy for cheap global detection,
- selective TMR/NMR only on the most critical modules,
- temporal retry only on detected failures,
- optional cold spares for lifetime reliability.